

1. a

$$\oint H \cdot dl = H \cdot L$$

$$n = \frac{N}{L_s}$$

$$I_{\text{ingesloten}} = nL \cdot I$$

$$H \cdot L = nL \cdot I$$

$$H = nI = \frac{N}{L_s} I$$

b. $B = \mu H$

$$= \mu_0 \mu_r H$$

$$B(z_p) = \mu_0 H \left(1 + (\mu_r - 1) \frac{z_p}{L_s} \right)$$

$$= \mu_0 \frac{N}{L_s} I \left(1 + (\mu_r - 1) \frac{z_p}{L_s} \right)$$

c. Nee, want de overgang tussen de velden moet continu zijn,

toch $B_{\text{lucht}} \neq B_{\text{materiaal}}$. Dat is fysisch niet correct.

Endat komt omdat we zwaar vereenvoudigde modellen gebruiken.

$$2. W = \frac{1}{2} \int_V H \cdot B dV$$

$$A = \pi r_s^2$$

$$\hookrightarrow \begin{cases} W_{\text{Luft}} = \frac{1}{2} H \cdot (\mu_0 H) \cdot A (L_s - z_p) \\ = \frac{1}{2} \mu_0 H^2 A (L_s - z_p) \end{cases}$$

$$\begin{cases} W_{\text{Platz}} = \frac{1}{2} H \cdot (\mu_0 \mu_r H) \cdot A_{z_p} \\ = \frac{1}{2} \mu_0 \mu_r H^2 A_{z_p} \end{cases}$$

totale energie

$$W(z_p) = \frac{1}{2} \mu_0 H^2 A (L_s + (\mu_r - 1) z_p)$$

$$(H = \frac{N}{L_s} I) \rightarrow$$

$$W(z_p) = \frac{1}{2} \mu_0 \frac{N^2 I^2}{L_s^2} A (L_s + (\mu_r - 1) z_p)$$

3.

$$F = \left| \frac{dW}{dz_p} \right|$$

$$W(z_p) = \frac{1}{2} \mu_0 \frac{N^2 I^2}{L_s} A (L_s + (\mu_r - 1) z_p)$$

$$\frac{dW}{dz_p} = \frac{1}{2} \mu_0 \frac{N^2 I^2}{L_s} A (\mu_r - 1)$$

$$A = \pi r_s^2 \rightarrow$$

$$F = \frac{1}{2} \mu_0 \pi r_s^2 \frac{N^2 I^2}{L_s} (\mu_r - 1)$$

